

Hotel Network Architecture Design

Comprehensive Laboratory Project Report

Course Title: Computer Networks Lab

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1 Project Introduction & Topology Overview

This report documents the design and implementation of a professional network architecture for a hotel enterprise system. The complete network has been modeled, configured, and verified using Cisco Packet Tracer. The design features end-to-end connectivity, clear departmental isolation, multi-protocol dynamic routing with path redundancy, secure wireless mobility, and unified application services.

1.1 Network Topology Screenshot

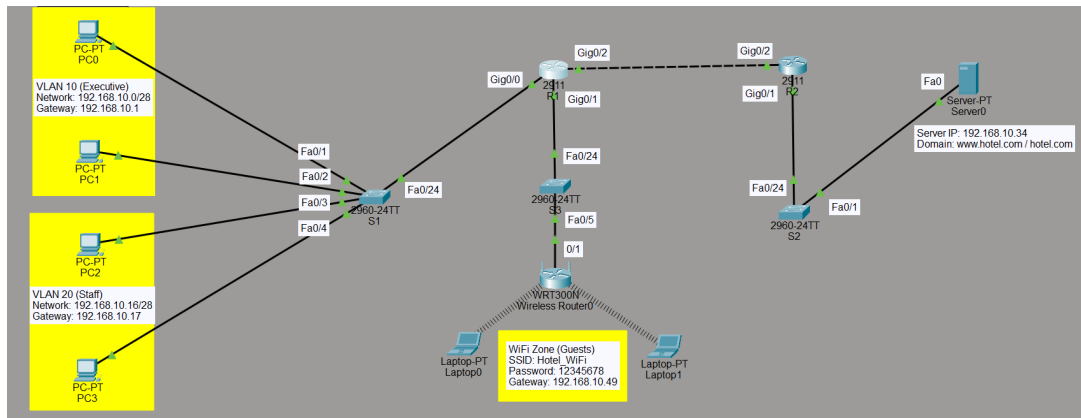


Figure 1: Complete Network Topology Layout in Cisco Packet Tracer.

2 Subnetting and Variable Length Subnet Masking (VLSM)

2.1 Why Subnetting and VLSM are Used

Subnetting is used to divide a large, single block of IP addresses into smaller, manageable sub-networks. In a default network, all devices belong to one large broadcast domain, causing huge packet traffic jams.

Variable Length Subnet Masking (VLSM) is better than Fixed Length Subnet Masking (FLSM) because FLSM forces every subnet to be the exact same size, which wastes hundreds of valid IP addresses. VLSM allows us to allocate custom-sized address spaces based on the exact requirements of each individual department.

2.2 Advantages and Disadvantages of VLSM

Advantages:

- **No Address Waste:** Maximum utilization of IP address space.
- **Flexible Sizing:** Subnets can be sized uniquely for different departments.
- **Better Security:** Different subnets restrict unauthorized inter-subnet snooping.

Disadvantages:

- **Complex Management:** Harder to track and design manually.
- **Configuration Overhead:** Requires sub-interfaces and complex routing setups.

2.3 Problems If VLSM Is Not Used

Without VLSM, if we allocate a default Class C network (/24) to a section requiring only 10 hosts, 244 IP addresses are permanently lost and wasted. This leads to quick IP exhaustion across enterprise projects.

2.4 Subnet Calculations and VLSM Table

Our base address is 192.168.10.0. To support up to 14 usable hosts per department block, we use a Custom Subnet Mask of 255.255.255.240 (/28 prefix).

Table 1: Official Network VLSM Planning Map

Segment Description	Network IP	Subnet Mask	Gateway IP	VLAN ID
VLAN 10 (Executive Block)	192.168.10.0	255.255.255.240	192.168.10.1	10
VLAN 20 (Staff Block)	192.168.10.16	255.255.255.240	192.168.10.17	20
Centralized Server Zone	192.168.10.32	255.255.255.240	192.168.10.33	–
Wireless Local Guest Zone	192.168.10.48	255.255.255.240	192.168.10.49	–

3 IP Assignment and Server Configurations

3.1 Why DHCP is Used Over Static IP

Static IP configuration requires typing individual IPs on every endpoint manually. In a dynamic hotel where guests arrive and leave constantly, static IP assignment is impossible to maintain. We implemented a centralized DHCP Server to auto-provision all client IP profiles.

3.2 Advantages of DHCP and Disadvantages of Static IP

Disadvantages of Static IP:

- Very slow deployment times.
- High chance of human typos causing duplicate IP conflicts.

Advantages of DHCP:

- Zero manual configuration on user computers.
- Automatic recycling of IP addresses when users disconnect.

3.3 Server Configuration Details

A single enterprise Server (**Server0**) was successfully configured with the following active services:

1. **DHCP Server:** Contains 3 pools (VLAN10.Pool, VLAN20.Pool, WiFi.Pool) to auto-lease network configurations.
2. **DNS Server:** Maps domain query string `www.hotel.com` directly to the server host IP address `192.168.10.34`.
3. **HTTP Server:** Serves hyper-text landing pages for client web browsers.
4. **Email Server (SMTP/POP3):** Handles hotel mail traffic via the `hotel.com` internal email domain.

3.4 Server Testing Verification Screenshots



Figure 2: Web Browser Resolution Test.

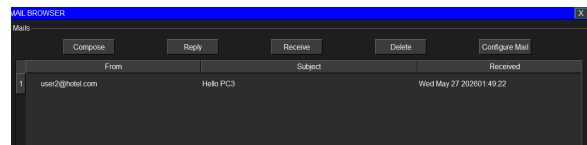


Figure 3: Email Delivery Status Log.

Figure 4: Successful Enterprise Server Module Verification Results.

4 VLAN Segmentation and Inter-VLAN Routing

4.1 Broadcast Domains Explained

A broadcast domain is a logical segment of a network where any device can send a data packet that is received by all other devices on that same segment. As we add more switches to a flat network, the physical size of the broadcast domain grows larger. This means broadcast storms can slow down the entire system.

4.2 Why VLANs are Used

Virtual Local Area Networks (VLANs) split a single physical switch into multiple virtual switches.

- **VLAN Advantages:** Stops broadcast storms from spreading, reduces background traffic noise, and secures sensitive departmental segments.
- **VLAN Disadvantages:** Requires specialized Layer-3 devices or sub-interface routers to cross boundaries.

4.3 Inter-VLAN Routing and Router-on-a-Stick

Devices in different VLANs cannot talk to each other by default. To enable communication, we configured **Router-on-a-Stick** by splitting the single physical router link into virtual sub-interfaces with IEEE 802.1Q encapsulation tags. Without this configuration, different departments would remain permanently isolated.

5 Router Interface Assignments and Dynamic Routing

5.1 Router Interface IP Mapping

The table below logs the physical port configurations on the core network router.

Table 2: Router R1 Full Interface Configuration Logs

Router Physical/Virtual Interface	Assigned IP Address	Network Subnet Mask
GigabitEthernet0/0.10 (VLAN 10 Sub)	192.168.10.1	255.255.255.240
GigabitEthernet0/0.20 (VLAN 20 Sub)	192.168.10.17	255.255.255.240
GigabitEthernet0/0 (Main Interface)	– (No IP Address)	–
GigabitEthernet0/1 (To Wi-Fi Hub)	192.168.10.49	255.255.255.240

5.2 Why Dynamic Routing is Used over Static Routing

Static routing forces the administrator to manually write down every network path inside the routing table. If any physical line drops or changes, static routing fails instantly until manual intervention. Dynamic routing protocols allow routers to automatically share paths and instantly route around broken lines.

5.3 OSPF vs RIP Comparison

Our network uses both protocols simultaneously to guarantee full compatibility and testing validation:

- **OSPF (Open Shortest Path First):** A fast, modern Link-State routing protocol. It uses cost metrics based on bandwidth and converges almost instantly, making it highly preferred for modern corporate networks.
- **RIPv2 (Routing Information Protocol v2):** A simpler Distance-Vector protocol limited by a maximum length of 15 router hops. It is lightweight but updates slowly compared to OSPF.

6 Network Security via Access Control Lists (ACL)

6.1 Security Purpose and ACL Rules

To prevent unauthorized entry into server configurations, we deployed an Extended IP Access Control List (ACL 100) directly onto the inbound pipeline of the Staff segment interface.

6.2 ACL Rules Configuration

The policy states that the Staff block (192.168.10.16/28) is blocked from accessing Web Services (TCP Port 80) on the core Server host machine (192.168.10.34), while all alternate nodes remain fully permitted to link normally.

```
1 access-list 100 deny tcp 192.168.10.16 0.0.0.15 host
   192.168.10.34 eq www
2 access-list 100 permit ip any any
3 interface gigabitEthernet 0/0.20
4 ip access-group 100 in
```

7 Wireless Mobility Configuration

To provide secure network access for guests and roaming users, a physical Home Wireless Router device was configured as a transparent access hub.

- **Network Port Mapping:** Core switch uplink line is routed into the local **LAN (Ethernet 1)** port instead of the isolated Internet port to share the same local network subnet pool.
- **SSID Broadcast Identifier:** Hotel_WiFi
- **Wireless Security Protocol:** WPA2-Personal using AES encryption keys.
- **Passkey Credentials:** 12345678
- **Client Mobility:** Laptops were swapped with wireless antennas and pulled IP dynamic settings directly from the main server DHCP pool.

8 Discussion

8.1 Limitations of the Current Network

While this network design satisfies all core project criteria, it has two key limitations:

1. **Single Server Bottleneck:** The HTTP, DNS, DHCP, and Email engines are running on one physical server node. If this single server crashes, all core network services stop working instantly.
2. **Lack of Edge Firewall Integration:** The setup relies purely on basic router ACL rules instead of using a dedicated edge security appliance (like an ASA Firewall).

8.2 Future Improvements

To expand this setup into a more robust and professional environment, we plan to implement the following changes in future designs:

1. **Server Redundancy:** Deploy secondary backup servers running distinct DHCP failover scopes and DNS secondary zones to prevent downtime.
2. **NAT Configuration:** Integrate Network Address Translation (NAT) rules to mask our internal corporate address spaces behind secure public IP configurations.
3. **Layer-3 Core Switching:** Upgrade from a single Router-on-a-Stick setup to dedicated Core Layer-3 Switches to improve performance.